

High extinction ratio and low loss polarization beam splitter based on multimode interference for PICs

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A polarization beam splitter (PBS) with a high extinction ratio that is based on multimode interference (MMI) is proposed and experimentally demonstrated on a silicon-on-insulator platform. The PBS is designed using cascaded MMI to enhance the extinction ratio (ER) and minimize insertion loss (IL) compared to a conventional MMI structure. The proposed device has a high extinction ratio (22.3 dB for TE and 25.6 dB for TM) and a low insertion loss (0.7 dB for TE and 0.1 dB for TM) at a wavelength of 1550 nm while the device supports a broad bandwidth of 60 nm (ER > 16 dB) for both polarizations over a wavelength between 1520–1580 nm. Moreover, the eigenmode expansion method shows that the device has a large fabrication tolerance of an MMI width ± 40 nm and a length ± 1500 nm. The proposed PBS can be used in on-chip silicon photonic integrated circuits for polarization division multiplexing. © 2020 Optical Society of America

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1. INTRODUCTION

Polarization division multiplexing (PDM) is a promising technology to enhance the bandwidth capacity of optical communication systems by transmitting two independent optical signals simultaneously having same wavelength at different polarization states [1]. The polarization manipulation is a key factor in photonic integrated circuits (PICs). In a silicon-on-insulator (SOI) platform, polarization sensitivity is a major issue since a SOI waveguide has high birefringence due to a submicron size and high refractive index contrast [2]. The polarization beam splitter (PBS) is a fundamental element for the realization of PDM [3] and a WDM receiver [4]. High-performance PICs based on PDM require the PBS to have desirable features such as a low insertion loss (IL), high extinction ratio, and a broad bandwidth. Various techniques have been reported for the design of an integrated PBS based on different structures, including a Mach–Zehnder interferometer [5], a directional coupler [6,7], a multimode interference coupler [8–10], an adiabatic coupler [11], hybrid plasmonic couplers [12], and a slanted silicon waveguide [13]. Compared to other structures, the MMI structure has an advantage because of its low excess loss, large fabrication tolerances, and ease of fabrication [14]. A PBS based on a directional coupler [15], subwavelength grating [16], and plasmonic waveguide [17] requires very precise control of the parameters due to high fabrication and temperature sensitivity. A PBS based on a conventional MMI requires a large device length since it uses the self-imaging principle to find the common multiples

beat length of TE and TM polarizations. Subsequently, the quasi-state imaging [18] characteristics of a MMI coupler are employed to minimize the length of the device, but it leads to a high insertion loss. The cascaded MMI-based PBS [19,20] has been used to minimize the insertion loss of the device. However, there is still a possibility to further improve the PBS in terms of the insertion loss, extinction ratio, and bandwidth.

In this paper, a PBS based on a cascaded MMI coupler is proposed and experimentally demonstrated on SOI. First, we have designed a PBS based on a conventional MMI structure to separate the TE and TM polarization input. In this structure, the quasi-state imaging principle is employed to find the beat length of the TE and TM polarizations and then the TM input is separated at the cross port due to a shorter beat length. Subsequently, we have modified the structure into a cascaded MMI to separate the TE polarization input. The beat length of the TE and TM mode is calculated analytically and the length of the device is optimized using eigenmode expansion method. The proposed device has a high extinction ratio (22.3 dB for TE and 25.6 dB for TM) and a low insertion loss (0.7 dB for TE and 0.1 dB for TM) at an operating wavelength 1550 nm. The device has a compact footprint ($3 \times 102.1 \mu\text{m}^2$) with a high fabrication tolerance. The on-chip grating couplers are typically used at the input and output port to characterize fabricated photonic devices [21,22]. The proposed PBS can be used in PICs for PDM.

2. DEVICE STRUCTURE AND SIMULATION

The PBS-based multimode interference coupler works on the principle of self-imaging [14]. According to this principle, the input mode is decomposed to various eigenmodes in the MMI region and they interfere with each other along the propagation direction due to different propagation constants. In a conventional MMI-based PBS, TE and TM polarization input can be separated into the bar and cross port by satisfying

$$L_{\text{MMI}} = m(3L_{\pi\text{TE}}) = n(3L_{\pi\text{TM}}), \quad (1)$$

where m and n are even and odd integers, respectively, or vice versa. For general interference in the MMI, the first self-imaging occurs at length $3L_{\pi}$. The beat length (L_{π}) can be calculated by

$$L_{\pi} = \frac{\pi}{\beta_0 - \beta_1} \approx 4n_{\text{eff}}^2 W_{\text{MMI}}^2 / 3\lambda, \quad (2)$$

where n_{eff} and W_{MMI} are the effective index and width of the MMI coupler, respectively; λ is the operating wavelength; and β_0 and β_1 are the propagation constants of the two lowest modes and can be calculated as

$$\Delta\beta = \beta_0 - \beta_p = p(p+2)\pi/3L_{\pi}. \quad (3)$$

The quasi-state image principle [18] can be approximated for the four lowest modes from the self-imaging principle. The length of the quasi-state image of the MMI is calculated by

$$L_{\text{MMI}} = 3(Q \pm \frac{1}{5})L_{\pi}, \quad (4)$$

where Q is an integer; thus, the length of the MMI can be reduced to one-fifth of the length obtained from perfect self-imaging for the TE and TM polarization.

A. PBS Based on Conventional MMI Structure

The schematic diagram of a PBS based on a conventional multimode interference structure is shown in Fig. 1. The structure is composed of a multimode section and an input/output tapered waveguide to reduce modal mismatch loss. The conventional MMI-based PBS has an optimized multimode width of $W_{\text{MMI}} = 3 \mu\text{m}$ and a length of $L_{\text{MMI}} = 52.3 \mu\text{m}$. The input and output port have tapered waveguides with widths $W_s = 0.5 \mu\text{m}$ and $W_t = 1 \mu\text{m}$, and a length $L_t = 10 \mu\text{m}$. The cross-section of the device consists of a buried SiO₂ substrate, silicon core, and a SiO₂ cover with thicknesses of $2 \mu\text{m}$, 220 nm , and $2.2 \mu\text{m}$, respectively.

The proposed polarization beam splitter is designed and analyzed using the three-dimensional (3D) eigenmode expansion method [23]. The EME method is used since it considers the

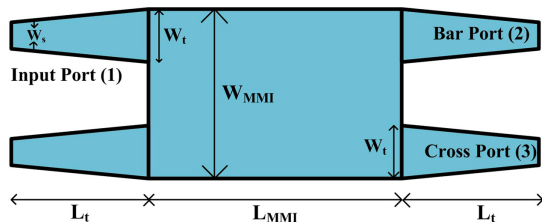


Fig. 1. Schematic diagram of a PBS based on a conventional MMI structure.

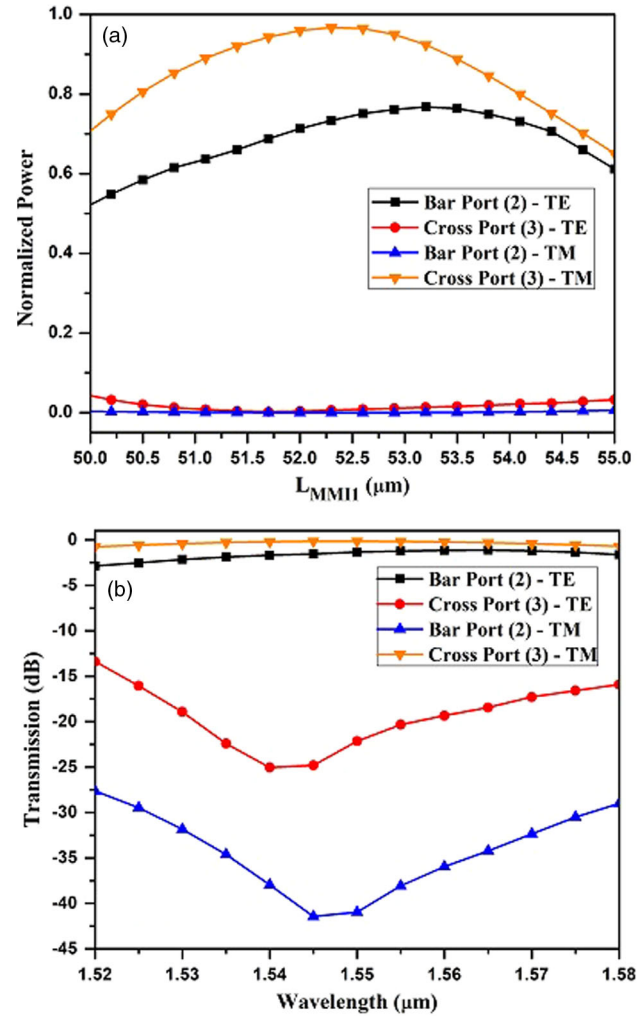


Fig. 2. (a) Normalized power with L_{MMI} and (b) Transmission power at ports 2 and 3 as a function of wavelength for TE and TM polarization.

vectorial properties of light for high index contrast of the SOI waveguide. In this device, the quasi-state imaging principle is employed to minimize the beat length of the MMI structure. The length of device is calculated from the beat length of the

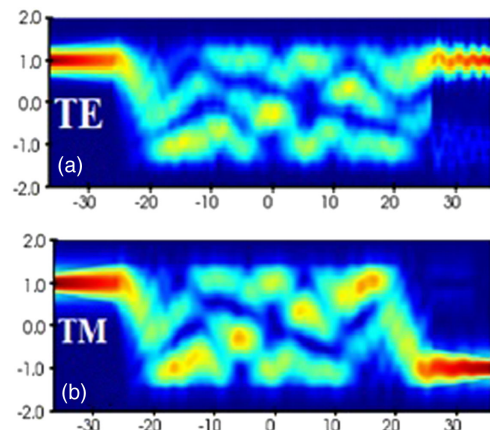


Fig. 3. Simulated optical field profile evolution of a PBS for (a) TE mode and (b) TM mode.

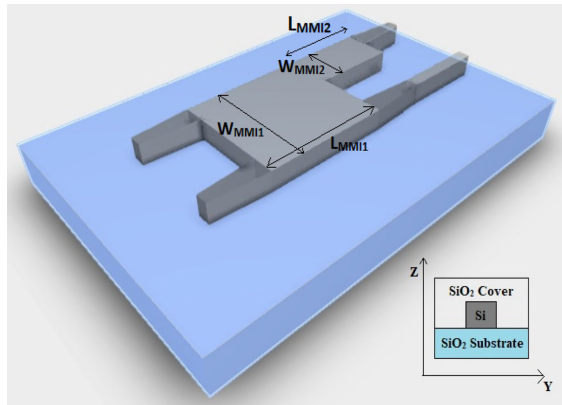


Fig. 4. Schematic diagram of the proposed PBS based on cascaded MMI.

MMI structure and optimized using the EME method for TE and TM polarization. The length of the PBS is optimized in such way that the TE and TM input are separated at the bar port (2) and cross port (3), respectively. The variation of the normalized power of the device with the MMI length at a wavelength

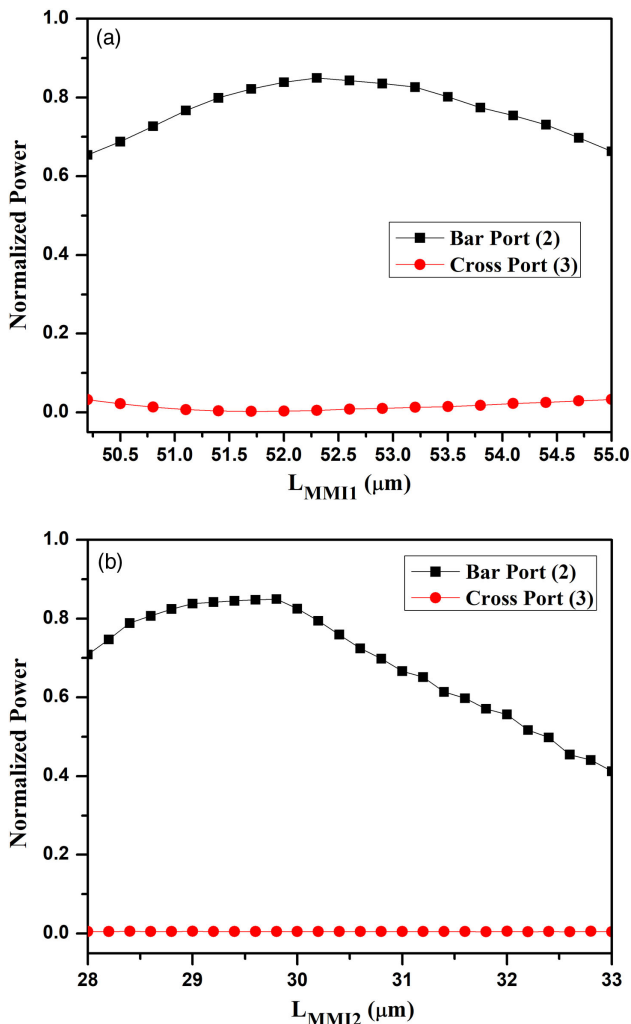


Fig. 5. (a) Normalized power for different L_{MMI1} and (b) for different L_{MMI2} for TE mode.

of 1550 nm is shown in Fig. 2(a). The figure shows that the normalized power is maximum for the TM input. So, it is separated from the cross port at the MMI length $L_{\text{MMI1}} = 52.3 \mu\text{m}$, and the output for TE input is separated from the bar port having a higher insertion loss that can be reduced by the implementation of a cascaded MMI. The transmission characteristics of this conventional MMI-based PBS for a wavelength between 1520–1580 nm are shown in Fig. 2(b).

The simulated optical field evolution of the conventional MMI-based PBS for the TE and TM modes at a wavelength of 1550 nm is shown in Fig. 3.

B. PBS Based on Cascaded MMI

The schematic diagram of the proposed cascaded MMI-based PBS is shown in Fig. 4. It consists of two MMIs (cascaded structure) that are used to separate the TM mode at the output port of the first MMI coupler and the TE mode at the second MMI coupler. The structure is modified to enhance the extinction ratio and minimize the insertion loss for the TE/TM polarization. The dimensions of first MMI coupler are the same as depicted in the above conventional MMI structure and the second MMI

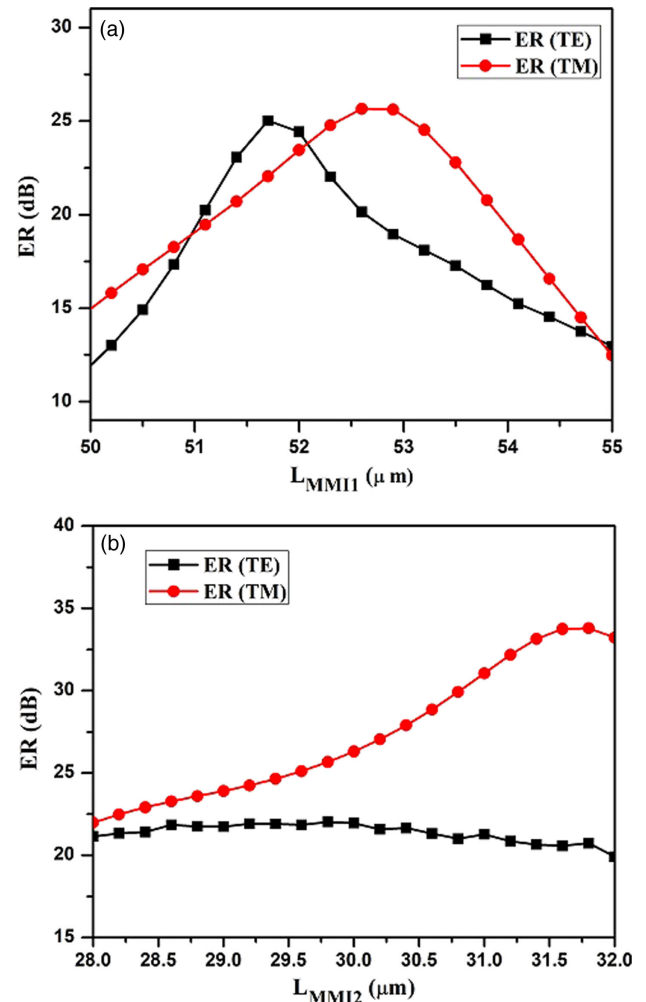


Fig. 6. (a) Extinction ratio versus L_{MMI1} and (b) Extinction ratio versus L_{MMI2} for TE and TM polarization.

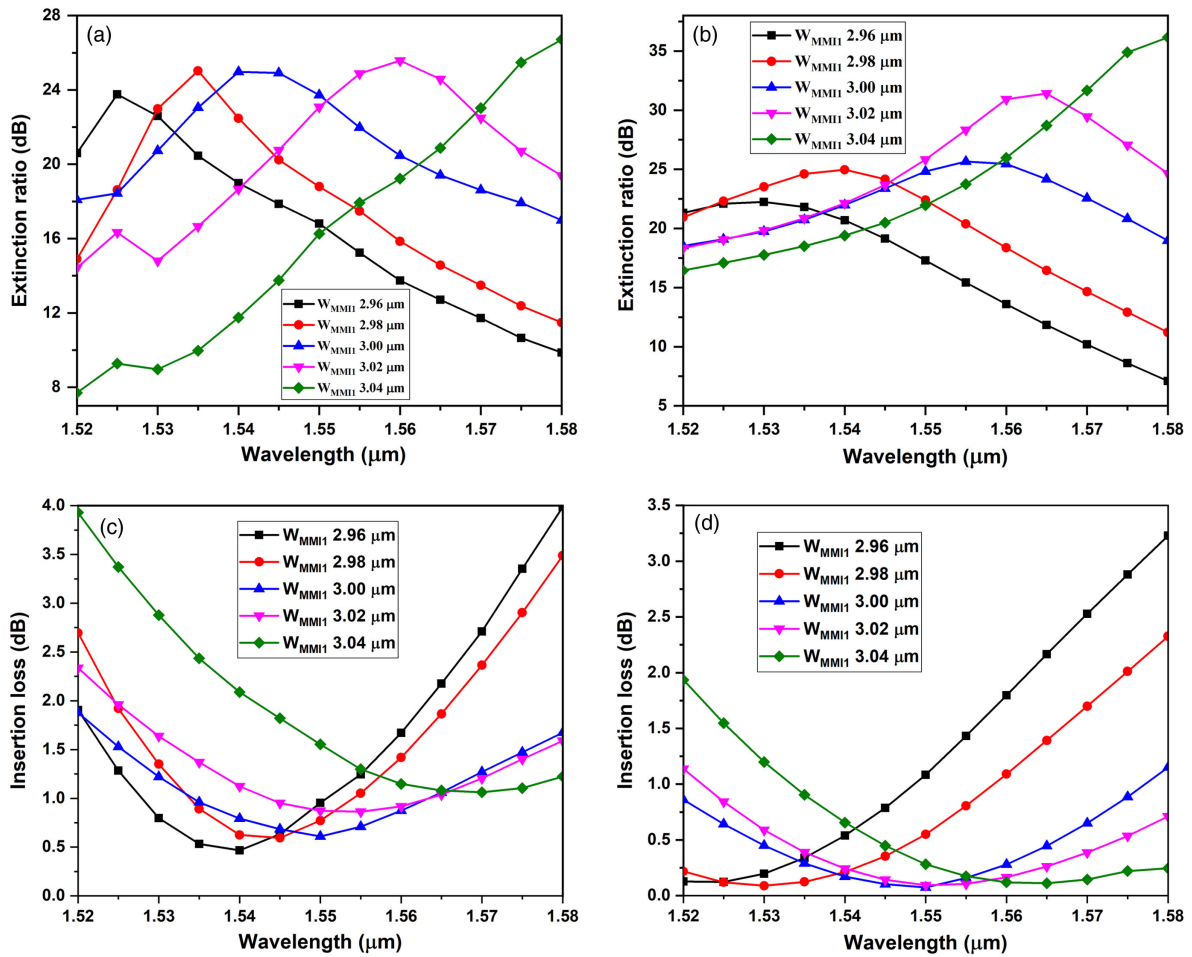


Fig. 7. Extinction ratio versus wavelength for different values of W_{MMI1} for (a) TE mode, (b) TM mode, (c) insertion loss versus wavelength for different values of W_{MMI1} for TE mode and (d) TM mode.

has an optimized width of $W_{MMI2} = 1.76 \mu\text{m}$ and a length of $L_{MMI2} = 29.8 \mu\text{m}$.

A finite difference eigenmode solver (FDE) and a full-vectorial 3D eigenmode expansion (EME) method are used to analyze the modal and propagation characteristics of the silicon-based optical devices, respectively. The variation of normalized power with different MMI lengths L_{MMI1} and L_{MMI2} for TE polarization is shown in Figs. 5(a) and 5(b), respectively. We can observe from the figure that the output normalized power for the TE mode is improved compared to a conventional MMI structure. The maximum output power is obtained at MMI lengths of $L_{MMI1} = 52.3 \mu\text{m}$ and $L_{MMI2} = 29.8 \mu\text{m}$.

The performance of the PBS is also analyzed in terms of the extinction ratio. The variation of extinction ratio for TE and TM polarizations with different MMI lengths is shown in Fig. 6. The figure shows that the extinction ratio is greater than 16 dB for a large variation in the MMI length. Thus, the fabrication tolerance of the proposed device in terms of length L_{MMI1} is $\pm 1500 \text{ nm}$, and for L_{MMI2} , it is $\pm 2000 \text{ nm}$.

The extinction ratio and the insertion loss of the device is also analyzed over a wavelength range of 1520–1580 nm by varying the width of the MMI for TE and TM polarization, as shown in Fig. 7. The performance of the proposed PBS is also

investigated for broad bandwidth applications. Figures 7(a) and 7(b) show the variation of the extinction ratio of the device with wavelength. The fabrication tolerance in terms of the MMI width (W_{MMI1}) is nearly $\pm 20 \text{ nm}$ for the entire C-band and has a better tolerance of $\pm 40 \text{ nm}$ at a wavelength of 1550 nm for

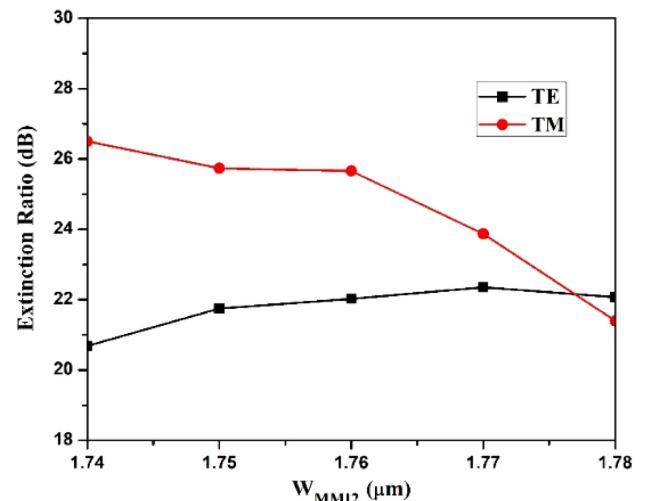


Fig. 8. Extinction ratio versus W_{MMI2} TE and TM polarization.

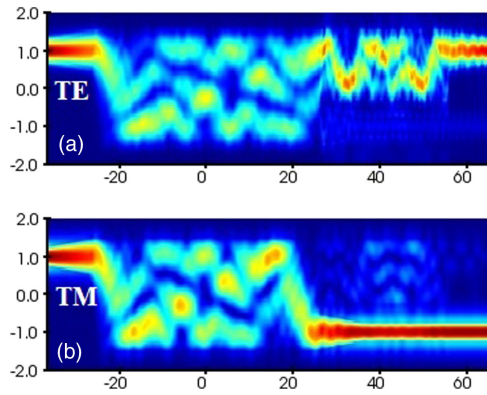


Fig. 9. Simulated optical field evolution of cascaded MMI-based PBS: (a) TE mode and (b) TM mode.

both TE and TM polarizations. It is observed that the proposed device has an extinction ratio of 22.3 dB for TE polarized input and 25.6 dB for TM polarized input at a wavelength of 1550 nm while the extinction ratio is >16 dB for both polarizations for a wavelength range of 1520–1580 nm. The insertion loss of the device with wavelength is shown in Figs. 7(c) and 7(d) for the TE and TM modes, respectively. The variation of insertion loss is minimum for the MMI width of $3\ \mu\text{m}$ over wavelengths between 1520–1580 nm.

The extinction ratio of the proposed PBS device is also analyzed by varying the width of the second MMI, which is shown in Fig. 8. From the figure, it is observed that the extinction ratio is optimum at a MMI width of ($W_{\text{MMI}2}$) = $1.76\ \mu\text{m}$. The simulated field evolution of the proposed PBS for TE and TM polarizations is shown in Figs. 9(a) and 9(b), respectively.

3. FABRICATION AND EXPERIMENTAL ANALYSIS

The PBS device has been fabricated using electron beam lithography (100 KeV) technology. A SOI wafer is used as the base material for fabrication, with a silicon thickness and a buffer oxide thickness of 220 nm and $2\ \mu\text{m}$, respectively. The substrate is spin-coated with hydrogen silsesquioxane (HSQ) as a resist. The PBS device is patterned using an electron beam instrument (Raith EBPG 5000+) with a raster step size of 5 nm. A buffer oxide etch is used to remove the resist from the surface and the device is inspected for the verification of patterning and the etch quality using a scanning electron microscope (SEM). The plasma-enhanced chemical vapor deposition (PECVD) technique is used to deposit a silicon dioxide (SiO_2) cladding layer of $2.2\ \mu\text{m}$ at 300°C . Reflectometry measurements are performed to verify the device layers and dimensions throughout the processes. The SEM image of the fabricated device is shown in Fig. 10.

The characterization of the device has been carried out using two test structures that consist of the proposed PBS connected to the input laser and output detectors through the input and output on-chip subwavelength grating couplers [21]. The two test structures are identical in all aspects, except the grating couplers. In the first test structure, TE polarized grating couplers are used. On the other hand, TM polarized grating couplers are

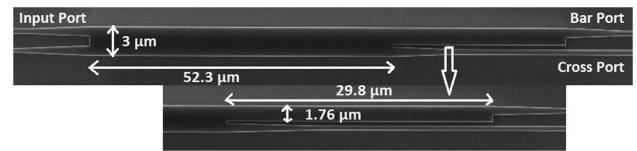


Fig. 10. SEM image of the fabricated PBS device.

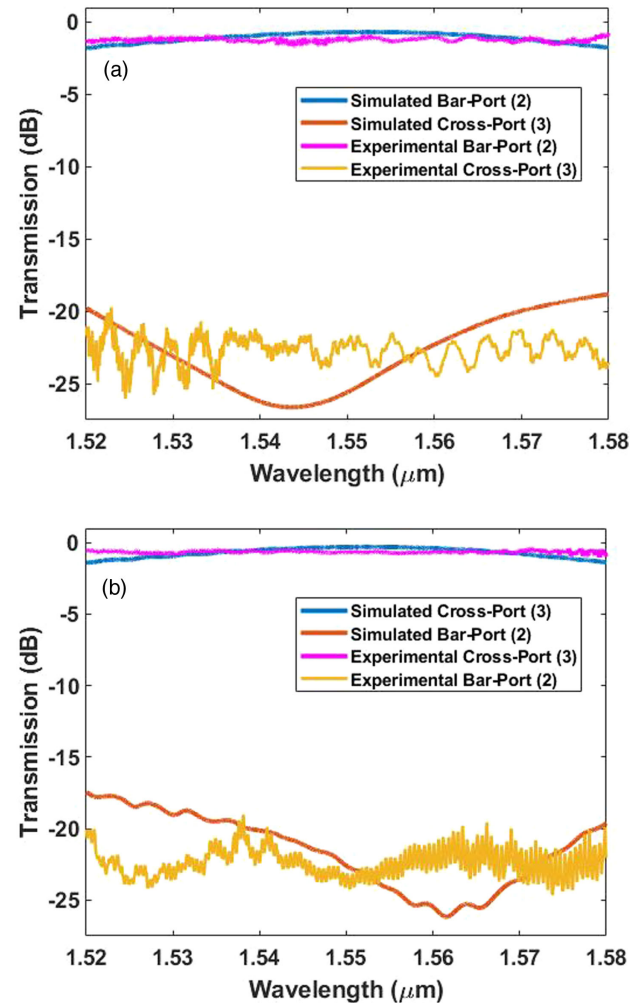


Fig. 11. Simulated and measured transmission spectrum of the proposed PBS for (a) TE and (b) TM mode.

used in the other test structure. The insertion loss of the input and output grating couplers has been calibrated out to find out the actual losses of the proposed PBS device. The transmission characteristics are analyzed for the wavelength range of 1520–1580 nm with 10 pm steps. The polarization state of the light is controlled by a polarization-maintaining fiber and the TE or TM polarizations are coupled into the grating couplers. The simulated and measured transmission characteristics of the proposed device for TE and TM polarizations are shown in the Fig. 11.

From Fig. 11, it is observed that the simulated transmission characteristics of the proposed PBS are similar to the experimental results for both TE and TM polarization with some fluctuations. The fluctuations can be due to the variation in

Table 1. Comparison of Previous PBS Designs to Our Proposed Device (ER Denotes Given Bandwidth of Device)

Structure	Footprint (μm^2)	IL (dB) TE	IL (dB) TM	Bandwidth (nm)	ER (dB) TE	ER (dB) TM	Fabrication Tolerance (nm)
Cascaded-MMI ^b [20]	7×600	3 at 1550 nm	2.1 at 1550 nm	25 for TE, 20 for TM	>20	>20	—
Angled MMI ^a [24]	25	0.38 at 1550 nm	0.79 at 1550 nm	53	>20	>20	NA
MMI ^b [25]	4.2×132.64	1.2	2.2	26 for TE, 55 for TM	>15	>15	ΔW : 90 ΔL : 5000
Cascaded-MMI ^b [26]	$L = 364$	0.9 at 1550 nm	2.8 at 1550 nm	55	>14	>20	NA
MMI-based SWG ^b [27]	4×92.7	1.9 at 1550 nm	2.5 at 1550 nm	84	>14.5	>11.7	NA
Asymmetrical DC ^a [28]	1.9×3.7	<1	<1	120	>14	>14	NA
Bent DC ^b [29]	1.5×1.3	<2	<1	40	15–16	15–20	ΔW : 36
Slot Waveguide ^a [30]	NA	<3.6	<0.16	37 for TE, 35 for TM	>31	>12	NA
Our Proposed Device ^b	3×102.1	0.7 at 1550 nm, <2 for 60 nm	0.1 at 1550 nm, <1.2 for 60 nm	60	22.3 at 1550 nm, >16	25.6 at 1550 nm, >16	ΔW : ± 40 ΔL : ± 1500

^aSimulated work.^bSimulated and experimental work.

the fabrication of the subwavelength grating coupler, which leads to a shift in the spectrum of the PBS device. The insertion loss (IL) of the proposed device is measured to be 0.7 dB and 0.1 dB for TE and TM, respectively, at a wavelength of 1550 nm, while IL is less than 2 dB for TE and 1.2 dB for TM for a wavelength range of 1520–1580 nm. A comparative analysis is performed among the various reported PBS and is shown in Table 1. The proposed cascaded MMI-based PBS has several advantages: a high extinction ratio, a low insertion loss, and a high fabrication tolerance. The PBS based on other structures (directional coupler, slot waveguide, subwavelength grating) has a better bandwidth, but it requires precise control due to high fabrication sensitivity.

4. CONCLUSION

A low loss polarization beam splitter with a high extinction ratio is proposed and experimentally demonstrated using a cascaded MMI. The device has a better performance compared to devices based on conventional MMI structures. The proposed PBS has a compact size and a low insertion loss of 0.7 dB and 0.1 dB for TE and TM polarizations, respectively, at a wavelength of 1550 nm. The device supports a broad bandwidth of 60 nm ($\text{ER} > 16$ dB) for both polarizations. Moreover, EME numerical tools show that the device has a high fabrication tolerance of a width ± 40 nm and a length ± 1500 nm. The proposed device can be used in silicon photonic integrated circuits for PDM.

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